

Whittaker Models and Wavefront Sets

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Preliminaries

F : $\mathbb{R}$ or (finite extensions of) p -adic fields

G : F -quasisplit group

$\mathfrak{g}$: Lie algebra of G

Let

$$\mathfrak{g} = \mathfrak{t} \oplus \bigoplus_{\alpha \in \Phi} \mathfrak{g}_{\alpha}$$

be the root space decomposition (with respect to a maximal torus T of G).

U_{α} : the root subgroup corresponding to $\alpha \in \Phi$.

Definition

Let (π, V_π) be a smooth representation of G , $B = TU$ a Borel subgroup of G defined over F , where T is a maximal torus contained in B , $U = R_u(B)$. Let ψ_U be a non-degenerate character of U . Define the space of Whittaker functionals of π associated to ψ_U as

$$Wh_{\psi_U}(\pi) := Hom_U(\pi, \psi_U).$$

We can call π having a (ordinary) Whittaker model if $\exists \psi_U$ such that

$$Wh_{\psi_U}(\pi) \neq 0.$$

We call a smooth representation π generic if π has a Whittaker model.

Whittaker Models

Whittaker Models correspond to regular nilpotent elements (orbits).

Definition

Let X be a nilpotent element in $\mathfrak{g}$. We call X regular if the centralizer $\mathfrak{g}^X$ of X in $\mathfrak{g}$ has dimension $\dim \mathfrak{g}^X = \text{rank}(\mathfrak{g})$.

Example: $G = GL_n$

Let B be the subgroup of lower triangular matrices, T the subgroup of diagonal matrices, $U = R_u(B)$. Fix an additive character $\psi^F : F \longrightarrow \mathbb{C}^\times$.

A standard non-degenerate character of U is

$$\psi_U(A) = \psi^F(a_{2,1} + a_{3,2} + \cdots + a_{n,n-1}), \quad A = (a_{ij})_{i,j} \in U$$

which is nontrivial on each simple root subgroup $1 + FE_{i+1,i}$.

On the Lie algebra level, by identifying $\mathfrak{g} \simeq \mathfrak{g}^*$, ψ_U corresponds to the Jordan block of size n .

Generalized Whittaker Models

G : connected classical group over F (or its metaplectic cover)

Pick a nilpotent element $X \in \mathfrak{g}$, inducing (by Jacobson-Morozov theorem) an $\mathfrak{sl}_2$ -triple $\{X, H, Y\}$. Then

$$\mathfrak{g} = \bigoplus_{i \in \mathbb{Z}} \mathfrak{g}_i^H$$

where

$$\mathfrak{g}_i^H := \{x \in \mathfrak{g} : \text{ad}(H)(x) = ix\}.$$

Put

$$\begin{aligned} u_X &:= \bigoplus_{i \leq -1} \mathfrak{g}_i^H, & U_X &:= \exp(u_X) \\ u_{X,2} &:= \bigoplus_{i \leq -2} \mathfrak{g}_i^H, & U_{X,2} &:= \exp(u_{X,2}) \end{aligned}$$

Generalized Whittaker Models

Fix an additive character ψ^F of F and a G -invariant non-degenerate bilinear form κ on $\mathfrak{g}$. Define ψ_X on $U_{X,2}$ by

$$\begin{aligned}\psi_X : U_{X,2} &\longrightarrow \mathbb{C}^\times \\ \exp(Z) &\longmapsto \psi^F(\kappa(X, Z))\end{aligned}$$

Let $(\omega_{\psi_X}, V_{\psi_X})$ be the smooth oscillator representation on U_X associated to ψ_X .

Note: If $\mathfrak{g}_{-1}^H = 0$, then $U_X = U_{X,2}$ and $\omega_{\psi_X} = \psi_X$.

Definition

Suppose (π, V_π) is a smooth representation of G , $X \in \mathfrak{g}$ a nilpotent element. Define the space of generalized Whittaker models of π associated to X as

$$Wh_X(\pi) := \text{Hom}_{U_X}(\pi, w_{\psi_X}).$$

Oscillator Representation

Definition of $(\omega_{\psi_X}, V_{\psi_X})$:

Let $\mathcal{H}_X := \mathfrak{g}_{-1}^H \times F$ be the Heisenberg subgroup with $\{0\} \times F$ being the center of $\mathcal{H}_X$. Consider

$$\begin{aligned}\alpha_X : U_X = \exp(\mathfrak{g}_{-1}^H)U_{X,2} &\longrightarrow \mathcal{H}_X \\ \exp(A)\exp(Z) &\longmapsto (A, \kappa(X, Z))\end{aligned}$$

Then we have $\psi_X = \psi^F \circ \alpha_X|_{U_{X,2}}$.

By Stone-von Neumann theorem, $\exists!$ irreducible unitarizable smooth representation $(\omega'_{\psi_X}, V_{\psi_X})$ of $\mathcal{H}_X$ with central character ψ^F . Define

$$\omega_{\psi_X} := \omega'_{\psi_X} \circ \alpha_X : U_X \longrightarrow GL(V_{\psi_X}).$$

Generalized Whittaker Models: Example

Example: $G = GL_n$

Let X be the (upper) Jordan block of size n . A standard choice of $\mathfrak{sl}_2$ -triple $\{X, H, Y\}$ is

$$H = \text{diag}\{n-1, n-3, \dots, -(n-1)\}, \quad Y = \sum_{i=1}^{n-1} i(n-i)E_{i+1,i}.$$

Then

$$\mathfrak{g} = \bigoplus_{k \in \mathbb{Z}} \mathfrak{g}_{2k}^H, \quad \text{where} \quad \mathfrak{g}_{2k}^H := \text{Span}_F\{E_{i,i+k} : 1 \leq i \leq n, 1 \leq i+k \leq n\}.$$

Example

Note that we have $\omega_{\psi_X} = \psi_X$ since $\mathfrak{g}_{-1}^H = 0$, and that

$$\mathfrak{u}_X = \bigoplus_{i \leq -1} \mathfrak{g}_i^H = \text{Span}_F\{E_{i,j} : i > j\}$$

is the subalgebra of strictly lower triangular matrices. We take $\kappa(A, B) := \text{tr}(AB)$ on $\mathfrak{g}$. Then for $Z = (z_{i,j}) \in \mathfrak{u}_X = \mathfrak{u}_{X,2}$,

$$\psi_X(\exp(Z)) = \psi^F(\kappa(X, Z)) = \psi^F(z_{2,1} + z_{3,2} + \cdots + z_{n,n-1}).$$

Therefore $\psi_X = \psi_U$. (Recall that ψ_U is the standard non-degenerate character of the subgroup U of lower triangular matrices.)

Definition

For a smooth representation π of G , define the Whittaker support of π as

$$WS(\pi) := \max_{\leq_F} \{O \in \mathcal{N}_F(\mathfrak{g})_\circ : Wh_X(\pi) \neq 0, X \in O\}.$$

Here $\mathcal{N}_F(\mathfrak{g})_\circ$ denotes the set of F -rational nilpotent orbits in $\mathfrak{g}$, and the maximality is taken under F -rational topological order.

Problem:

Which orbits appear in the Whittaker support (?)

Theorem

Let π be a smooth representation of G , $O \in WS(\pi)$. Then O must be a quasi-admissible orbit.

Admissibility of Orbits:

Definition

Suppose $\mathcal{O} \subseteq \mathfrak{g}$ is a nilpotent orbit, for $X \in \mathcal{O}$. For $X \in \mathcal{O}$, put

$$\text{Stab}(X) := \{g \in G : \text{Ad}(g)(X) = X\}.$$

Let

$$\widetilde{\text{Stab}(X)} \longrightarrow \text{Stab}(X)$$

be the metaplectic double cover. Call $\mathcal{O}$ admissible if, for some/every $X \in \mathcal{O}$, $\widetilde{\text{Stab}(X)} \longrightarrow \text{Stab}(X)$ splits over $(\text{Stab}(X))_0$. Call $\mathcal{O}$ quasi-admissible if $\widetilde{\text{Stab}(X)}$ admits a finite-dimensional genuine representation.

Nilpotent Orbits

Theorem

Let $\mathcal{O} \subseteq \mathfrak{g}$ be a nilpotent orbit.

- 1 If $G = GL_n(F)$ or $SL_n(F)$, then $\mathcal{O}$ is automatically admissible.
- 2 If $G = U(V)$ or $SU(V)$ for a Hermitian space V , then $\mathcal{O}$ is automatically quasi-admissible. If moreover F is p -adic, then $\mathcal{O}$ must be admissible.

Conjecture

When F is p -adic, for any connected classical group G (or its metaplectic cover), if $\mathcal{O} \subseteq \mathfrak{g}$ is quasi-admissible then it is admissible.

Nilpotent Orbits in $WS(\pi)$

Add more hypothesis on π :

Theorem

Suppose π is a smooth representation of G . Then,

- ① *if π is quasi-cuspidal, O must be F -distinguished if $O \in WS(\pi)$;*
- ② *if π is admissible and tempered, and either $F = \mathbb{R}$ or G classical, O must be F -distinguished if $O \in WF(\pi)$.*

Definition

Suppose $O \subseteq \mathfrak{g}$ is a nilpotent orbit. We say that O is F -distinguished if O do not intersect $\text{Lie}(L)$ for any proper Levi subgroup L of G .

Example: $G = GL_n$

The nilpotent orbit containing the Jordan block of size n is F -distinguished.

$$WS = WF$$

Suppose (π, V_π) is a smooth representation of G .

Theorem

If F is p -adic and G is classical, π is a smooth representation of finite length, then we have

$$WS(\pi) = WF(\pi)$$

where $WF(\pi)$ denotes the wavefront set of π .

Analytic Wavefront Set

$$M(G) : \begin{cases} \text{smooth representations of } G \text{ of finite length, if } F \text{ } p\text{-adic} \\ \text{admissible, finitely generated representations of } G, \text{ if } F = \mathbb{R} \end{cases}$$

Suppose $\pi \in M(G)$.

Let Θ_π be the distribution character of π defined by

$$\langle \Theta_\pi, f \rangle := \text{tr}(\pi(f)), \quad f \in C_c^\infty(G),$$

where

$$\pi(f) := \int_G f(g) \pi(g) dg.$$

Analytic Wavefront Set

By the Barbash-Vogan asymptotic expansion we have

$$\exp^*(\Theta_\pi) \sim \sum_{\mathcal{O}} c_{\mathcal{O}}(\pi) \widehat{\mu_{\mathcal{O}}}$$

near $0 \in \mathfrak{g}$. In particular, when F is p -adic, this becomes cleaner

$$\exp^*(\Theta_\pi) = \sum_{\mathcal{O}} c_{\mathcal{O}}(\pi) \widehat{\mu_{\mathcal{O}}}$$

in a sufficiently small neighborhood of 0.

Definition

Define the wavefront set of π as

$$WF(\pi) := \max_{\leq_F} \{ \mathcal{O} \in \mathcal{N}_F(\mathfrak{g})_{\circ} : c_{\mathcal{O}}(\pi) \neq 0 \}.$$

Analytic Wavefront Set

Let $\pi \in M(G)$ be an irreducible representation.

Analytic Wavefront Set:

Definition

Define the analytic wavefront set of π as

$$WF_{tr}(\pi) := \max_{\leq_{st}} \{O \in \mathcal{N}_F(\mathfrak{g})_{\circ} : c_O(\pi) \neq 0\}$$

where the maximality is taken under the F -stable topological order.

Algebraic Wavefront Set

Recall the Whittaker support of π :

$$WS(\pi) = \max_{\leq_F} \{O \in \mathcal{N}_F(\mathfrak{g})_{\circ} : Wh_X(\pi) \neq 0, X \in O\}.$$

Algebraic Wavefront Set:

Definition

Define the algebraic wavefront set of π as

$$WF_{wm}(\pi) := \max_{\leq_{st}} \{O \in \mathcal{N}_F(\mathfrak{g})_{\circ} : Wh_X(\pi) \neq 0, X \in O\}.$$

Arithmetic Wavefront Set

Suppose $\pi = \pi_a(\phi, \chi)$ via the Vogan Langlands correspondence ι_a , where (ϕ, χ) is an enhanced L -parameter of G .

By doing L -parameter descent [cf. Jiang, Liu, Zhang, 2025], we have the notion of **arithmetic wavefront sets**:

$$WF_{ari}(\pi) := \max_{\leq_{st}} WF_a(\phi, \chi).$$

As a corollary of $WS(\pi) = WF(\pi)$ when F is p -adic and G is classical, we already have

Theorem

If F is p -adic and G is classical, then

$$WF_{tr}(\pi) = WF_{wm}(\pi).$$

Wavefront Set Conjecture

Concerning wavefront sets from three different perspectives:

Conjecture

Let F be a local field of characteristic 0, $G = U_n(F), SO_n(F), Sp_{2n}(F), Mp_{2n}(F)$. Let ϕ be a generic local L -parameter of G . For any $\pi \in \Pi_\phi[G]$, we have

$$WF_{tr}(\pi) = WF_{wm}(\pi) = WF_{ari}(\pi).$$

$WF_{ari}(\pi)$ can often be computed explicitly:

Theorem

Suppose $F = \mathbb{R}$. Let ϕ be a local L -parameter of $G = U_n(F), SO_n(F), Sp_{2n}(F), Mp_{2n}(F)$ as above. Then, if $\widehat{\mathcal{S}}_\phi$ is trivial, then $WF_{ari}(\pi)$ consists of all regular nilpotent orbits; if $\widehat{\mathcal{S}}_\phi$ is nontrivial, then $WF_{ari}(\pi)$ contains only one orbit.

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Thank you!